

Linear Algebra Overview Part 2

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1 Introduction

Vector spaces, sometimes called linear spaces, are sets with some restrictions on what the elements (or objects) of the set can be. One such restriction is that the set is non-empty which simply forces that there are what we call "Vectors" in our vector space. The idea of what a vector can be is somewhat abstract, which allows for flexibility in mathematical ideas. This is because a vector space is defined from a set and just as a set can have elements that are abstract such as a set containing an apple, banana, and orange, so too can a vector space. In the case of the set call it A contains an apple, banana, and orange, can be used to build a vector space where we use the fact that these three elements are distinct and well defined as a foundation (or basis) to form linear combinations building a space of dimension 3.

2 Definitions and Theorems

Vector Space

A vector space is a non empty set V with two operations, addition and multiplication by scalars defined.

Further a vector space satisfy ten axioms. Let $u, v, w \in V$ and define the scalar field to be the real numbers $\mathbb{R}$.

Then if $c, d \in \mathbb{R}$ we have:

1. Closure under addition: $\vec{u} + \vec{v} \in V$
2. Commutativity: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3. Associativity: $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. The additive identity (zero vector) exists: $\vec{0} \in V, \vec{u} + \vec{0} = \vec{u}$
5. Additive inverse exists: $\exists (-\vec{u}) \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0}$
6. Closure under scalar multiplication: $c\vec{u} \in V$
7. Distributivity across vector addition: $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$
8. Distributivity across scalar addition: $(c + d)\vec{u} = c\vec{u} + d\vec{u}$
9. Associativity of scalar multiplication: $c(d\vec{u}) = (cd)\vec{u}$
10. The scalar multiplicative identity exists: $1\vec{u} = \vec{u}$

Note: Any field works for the definition of a vector space. But rather than defining the technical details of what a field is I present the real numbers which is a field, as is the complex numbers. For more information on fields refer to Wikipedia.

Further the real numbers $\mathbb{R}$ is not an algebraically closed field. This means there exist polynomials in $\mathbb{R}$ that do not have zeros in $\mathbb{R}$. An example being $y = x^2 + 1$ having complex zeros, $\pm i$. Algebraic closure is not required for a scalar field in the context of vector spaces but it is nice as it guarantees solutions. In practice I recommend using the complex numbers $\mathbb{C}$ as the scalar field on your vector spaces. But for these notes and lectures we use the real numbers as it simplify the definitions an some of the "rigor."

Subspace

A subspace H of a vector space V is a subset $H \subseteq V$ which has the following three properties

1. The additive identity is in H : $\vec{0} \in H$
2. Closure under addition: If $\vec{u}, \vec{v} \in H$ then $\vec{u} + \vec{v} \in H$
3. Closure under scalar multiplication: If $\vec{u} \in H$ and $c \in \mathbb{R}$ then $c\vec{u} \in H$

Linear Dependence

Let $V = \{\vec{v}_0, \dots, \vec{v}_n\}$ be an indexed set of at least two vectors with $\vec{v}_0 \neq \vec{0}$. Then the elements (vectors) in V are linearly dependent $\iff \vec{v}_i$ is a linear combination of the other elements in V for some $0 \leq i \leq n$. That is to say

$$\vec{v}_i = a_0 \vec{v}_0 + \dots + a_{i-1} \vec{v}_{i-1} + a_{i+1} \vec{v}_{i+1} + \dots + a_n \vec{v}_n$$

where $a_i \in \mathbb{R} \forall 0 \leq i \leq n$

Get this, the definition for linear independence is crazy!

Linear Independence

Let $V = \{\vec{v}_0, \dots, \vec{v}_n\}$ be an indexed set of at least two vectors with $\vec{v}_0 \neq \vec{0}$. Then the elements (vectors) in V are linearly independent if they are not linearly dependent.

3 Examples

3.1 vector space/subspace examples

Lets look at the set of all 2×1 matrices with real entry's. Call this set

$$V = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

Let's check that this is a vector space by checking the ten axioms of a vector space against the set V .

To start let's look at the first closure under addition. So let $a, b, c, d \in \mathbb{R}$

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d \end{bmatrix}$$

Since the sum of two real numbers is a real number we have $a + c \in \mathbb{R}$ and $b + d \in \mathbb{R}$ so we have closure under addition satisfying the first axiom. One down nine more to go!

Looking at the second axiom, we require commutativity. That is to say $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ So as $a, b, c, d \in \mathbb{R}$ we have

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d \end{bmatrix}$$

and

$$\begin{bmatrix} c \\ d \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} c + a \\ d + b \end{bmatrix}$$

and since $a, b, c, d \in \mathbb{R}$ we have $a + c = c + a$ and $b + d = d + b$ thus

$$\begin{bmatrix} a + c \\ b + d \end{bmatrix} = \begin{bmatrix} c + a \\ d + b \end{bmatrix}$$

which gives

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

So we have satisfied the second axiom, commutativity.

Moving on to the third axiom, associativity. This means we must satisfy $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$. We need more real numbers! Now let $a, b, c, d, m, n \in \mathbb{R}$

Then look at

$$\begin{aligned} \left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} \right) + \begin{bmatrix} m \\ n \end{bmatrix} &= \begin{bmatrix} a + c \\ b + d \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix} \\ &= \begin{bmatrix} a + c + m \\ b + d + n \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} \begin{bmatrix} a \\ b \end{bmatrix} + \left(\begin{bmatrix} c \\ d \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix} \right) &= \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c+m \\ d+n \end{bmatrix} \\ &= \begin{bmatrix} a+c+m \\ b+d+n \end{bmatrix} \end{aligned}$$

so we have

$$\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} \right) + \begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} + \left(\begin{bmatrix} c \\ d \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix} \right)$$

Thus we satisfy the third axiom of associativity! We really are off to a great start for this to be a vector space!

Lets keep going down the list to the fourth axiom, the additive identity, $\vec{0} \in V$, $\vec{u} + \vec{0} = \vec{u}$.

We know

$$\vec{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

but lets see if this in fact an additive identity.

$$\begin{aligned} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 0+a \\ 0+b \end{bmatrix} \\ &= \begin{bmatrix} a \\ b \end{bmatrix} \end{aligned}$$

as in the real numbers $0 + a = a$ and $0 + b = b$.

Thus we've satiated the fourth axiom of the additive identity as well.

Shall we attempt the fifth, additive inverse? Lets do it! We need $\exists (-\vec{u}) \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0}$. I've got a great guess for $-\vec{u}$, lets try

$$-\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -a \\ -b \end{bmatrix}$$

I'm feeling really good about this guess!

Then we have

$$\begin{aligned}\begin{bmatrix} a \\ b \end{bmatrix} + \left(-\begin{bmatrix} a \\ b \end{bmatrix}\right) &= \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} -a \\ -b \end{bmatrix} \\ &= \begin{bmatrix} a - a \\ b - b \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

Am I a good guesser or what! That worked! We have additive inverses satisfying the fifth axiom.

Looking at the sixth axiom, closure under scalar multiplication. Formally, $c\vec{u} \in V$. So lets now let $x, y \in \mathbb{R}$ in addition to $a, b, c, d, m, n \in \mathbb{R}$. Now we have

$$x \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} xa \\ xb \end{bmatrix}$$

and again as the multiplication of real numbers in a real number we have $c\vec{u} \in V$, so the sixth axiom of closure under scalar multiplication is upheld.

Considering the seventh axiom distributivity across vector addition, $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$. We have

$$\begin{aligned}
x\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}\right) &= x\begin{bmatrix} a+c \\ b+d \end{bmatrix} \\
&= \begin{bmatrix} x(a+c) \\ x(b+d) \end{bmatrix} \\
&= \begin{bmatrix} xa+xc \\ xb+xd \end{bmatrix} \\
&= \begin{bmatrix} xa \\ xb \end{bmatrix} + \begin{bmatrix} xc \\ xd \end{bmatrix} \\
&= x\begin{bmatrix} a \\ b \end{bmatrix} + x\begin{bmatrix} c \\ d \end{bmatrix}
\end{aligned}$$

So we have satisfied distributivity across vector addition.

Now to ponder the eighth axiom. Distributivity across scalars $(c+d)\vec{u} = c\vec{u} + d\vec{u}$. So let's now suppose $a, b, x, y \in \mathbb{R}$.

$$\begin{aligned}
(x+y)\begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} (x+y)a \\ (x+y)b \end{bmatrix} \\
&= \begin{bmatrix} xa+ya \\ xb+yb \end{bmatrix} \\
&= \begin{bmatrix} xa \\ xb \end{bmatrix} + \begin{bmatrix} ya \\ yb \end{bmatrix} \\
&= x\begin{bmatrix} a \\ b \end{bmatrix} + y\begin{bmatrix} a \\ b \end{bmatrix}
\end{aligned}$$

So we have the eighth axiom, distributivity across scalars.

Were so close! Taking a look at the ninth axiom, associativity of scalar multiplication $c(d\vec{u}) = (cd)\vec{u}$. We have

$$\begin{aligned}
x(y \begin{bmatrix} a \\ b \end{bmatrix}) &= x \begin{bmatrix} ya \\ yb \end{bmatrix} \\
&= \begin{bmatrix} x(ya) \\ x(yb) \end{bmatrix} \\
&= \begin{bmatrix} (xy)a \\ (xy)b \end{bmatrix} \\
&= (xy) \begin{bmatrix} a \\ b \end{bmatrix}
\end{aligned}$$

Giving us the ninth axiom, associativity of scalar multiplication.

To wrap up and conclude that the set of all 2×1 matrices with real entry's is a vector space we'll look at the tenth axiom, the scalar multiplicative identity exists $1\vec{u} = \vec{u}$. So we have

$$\begin{aligned}
1 \begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 1a \\ 1b \end{bmatrix} \\
&= \begin{bmatrix} a \\ b \end{bmatrix}
\end{aligned}$$

Which satisfy all ten of the axioms of a vector space forcing us to conclude that the set of all 2×1 matrices with real entry's is in fact a vector space!

Lets now look at some examples that break this, remembering that the criteria for a vector space is that it is built out of a set, thus any set satisfying the ten axioms is a vector space. Still the elements of sets can get weird.

One example of a set I love is the set of a marker, a pencil, and a pen. These three objects together for a set of three writing objects. There are no mathematical issues with this being a set. That said, it is not a vector space yet as I have not given a way to add the elements together or defined a scalar field.

If the reader is interested in practicing I recommend attempting to prove the set of all continuous functions is a vector space rather than my pen, pencil, marker example, and turning that into a vector space is fairly technical, but if you want try! As for the set of continues functions we will use this in our examples of subspaces. But I leave it to you

to prove it is a vector space itself.

Now lets look at stuff that's not a vector space! Our first example is the set of all natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$. Do make note that depending on the author and context sometime the natural numbers do not include 0. For our purposes lets assume 0 is a natural number as we need an additive identity! That's not the axiom I am going to break in this example. Our scalar field is going to be the real numbers as usual.

Rather than going through the axioms in order till one breaks were going to guess that the sixth axiom of closure under scalar multiplication $c\vec{u} \in V$ is going to break. Possibly because I made this problem to do that, but who's to say why this guess may or may not work?

So we need a counter example to show something is not a vector space. Given that our set is the natural numbers $\mathbb{N}$ we are looking at exclusively non-negative numbers. So we have 299,792,458 is a natural number, thus chose this random natural number in our counter example. Take our scalar we will multiply by to be -5, as -5 is a real number.

Then we have $-5 \times 299,792,458 = -1,498,962,290$ which is very much a negative number and thus not a natural number breaking the sixth axiom. This proves the set of natural numbers with scalars in the real numbers is NOT a vector space.

Lets now look at an example of a subspace. I am going to inaccurately assume you've proven that the set of continuous functions is a vector space. If you did, you're one of my favorite students! If you didn't you're still one of my favorite students. Anyways lets look at the set of polynomials, which is certainly a subset of the set of continuous functions. To start lets turn words into math symbols.

Say C is the set of continuous functions. Then

$$C = \{f : \mathbb{R} \rightarrow \mathbb{R} | f \text{ is continuous}\}$$

And say P is the set of polynomials. Then

$$P = \{p : \mathbb{R} \rightarrow \mathbb{R} | p = \sum_{i=0}^n a_i x^i, a_i \in \mathbb{R}, x \text{ is in the domain}\}$$

So as polynomials are continuous functions we have $P \subset C$. And since we proved C is a vector space lets see if P is a subspace.

To start lets see if the additive identity is in our subspace P , that is to say $\vec{0} \in P$. Well since

$$P = \{p : \mathbb{R} \rightarrow \mathbb{R} | p = \sum_{i=0}^n a_i x^i, a_i \in \mathbb{R}, x \text{ is in the domain}\}$$

We have

$$p_0 = \sum_{i=0}^0 b_i x^i, \text{ where } b_0 = 0$$

is an element of P . Let's try adding it to another element in P to see if it is really the additive identity.

say

$$p_n = \sum_{i=0}^n a_i x^i, \text{ where } a_i \in \mathbb{R}$$

then

$$\begin{aligned} p_0 + p_n &= \sum_{i=0}^0 b_i x^i + \sum_{i=0}^n a_i x^i \\ &= 0 + \sum_{i=0}^n a_i x^i \\ &= \sum_{i=0}^n a_i x^i \\ &= p_n \end{aligned}$$

So p_0 is the additive identity and is in P .

Let's now check to see if P is closed under addition. Which is to say $\vec{u} + \vec{v} \in H$. So take two polynomials one of order n and one of order m ,

$$p_n = \sum_{i=0}^n a_i x^i, \text{ where } a_i \in \mathbb{R}$$

and

$$p_m = \sum_{i=0}^m b_i x^i, \text{ where } b_i \in \mathbb{R}$$

Adding them together we get

$$p_n + p_m = \sum_{i=0}^n a_i x^i + \sum_{i=0}^m b_i x^i$$

Suppose without loss of generality $m \geq n$ and that $a_i = 0, \forall i > n$. So we have

$$\begin{aligned} p_n + p_m &= \sum_{i=0}^m a_i x^i + b_i x^i \\ &= \sum_{i=0}^m (a_i + b_i) x^i \end{aligned}$$

And since the sum of two real numbers is a real number we have demonstrated closure under vector addition.

To complete the verification that P being the set of all polynomials is a subspace of the vector space of continuous functions we need to check that we are closed under scalar multiplication.

Uhh, take p_n and multiply by, I don't know, the real number $o \in \mathbb{R}$. That should probably work. Note this is the letter o as a variable not a zero! Blame the guy I asked for a letter suggestion.

$$\begin{aligned} op_n &= o \sum_{i=0}^n a_i x^i \\ &= \sum_{i=0}^n oa_i x^i \end{aligned}$$

And $oa_i \in \mathbb{R}$ so were chilling and the set of polynomials is definitely a subspace of the vector space of continuous functions.

3.2 linear independence dependence examples

Turning our attention to linear dependence and independence.

Lets consider the our favorite example from part 1 and find out weather it is linearly dependent or independent.

$$\begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix} x + \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix} y + \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix} z = \begin{bmatrix} \ln(\pi) \\ 0 \\ 6 \end{bmatrix}$$

Where x, y, z are scaling the respective vectors they are attached to.

Rather than considering the linear combination as that does not make sense weather or not it is independent or

dependent lets look at the three matrices weighted by x, y, z . (Hint we already know its status considering linear independence from the information from part 1. I'll explain why we know after checking.)

So we have the set of vectors

$$\left\{ \begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix}, \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix} \right\}$$

To check if they are linearly dependent we want to write one of the matrices as a linear combination of the others. That is to say

$$x \begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix} + y \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix}$$

Or equivalently

$$x \begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix} + y \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix} + z \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This is nice as we can turn this into a systems of equations. If a solution exists other than the trivial solution being $x = y = z = 0$ then the set is linearly dependent.

To being solving lets rewrite our linear combination as a system of linear equations.

$$42x + 67y + 73z = 0$$

$$\sin(4)x - y + 0z = 0$$

$$-x + 2y + 24z = 0$$

Simplifying before starting

$$42x + 67y + 73z = 0$$

$$\sin(4)x - y = 0$$

$$-x + 2y + 24z = 0$$

For the funnies, lets start with the third equation this time.

$$-x + 2y + 24z = 0$$

$$\implies x = 2y + 24z$$

Going up the list of equations to the second

$$\sin(4)x - y = 0$$

$$y = \sin(4)x$$

$$y = \sin(4)(2y + 24z)$$

$$y = \sin(4)2y + \sin(4)24z$$

$$y - 2\sin(4)y = 24\sin(4)z$$

$$(1 - 2\sin(4))y = 24\sin(4)z$$

$$y = \frac{24\sin(4)}{1 - 2\sin(4)}z$$

Plugging into the top equation

$$42x + 67y + 73z = 0$$

$$\implies 42(2y + 24z) + 67\left(\frac{24\sin(4)}{1 - 2\sin(4)}z\right) + 73z = 0$$

$$\implies 42\left(2\frac{24\sin(4)}{1 - 2\sin(4)}z + 24z\right) + 67\left(\frac{24\sin(4)}{1 - 2\sin(4)}z\right) + 73z = 0$$

Lets solve for z ! And great news z factors out!

$$\begin{aligned}
 42\left(2\frac{24\sin(4)}{1-2\sin(4)}z + 24z\right) + 67\left(\frac{24\sin(4)}{1-2\sin(4)}z\right) + 73z &= 0 \\
 \left(42\left(2\frac{24\sin(4)}{1-2\sin(4)} + 24\right) + 67\left(\frac{24\sin(4)}{1-2\sin(4)}\right) + 73\right)z &= 0 \\
 \implies z &= 0
 \end{aligned}$$

As

$$42\left(2\frac{24\sin(4)}{1-2\sin(4)} + 24\right) + 67\left(\frac{24\sin(4)}{1-2\sin(4)}\right) + 73 \neq 0$$

So lets plug z into the second equation to solve for y !

$$\begin{aligned}
 y &= \frac{24\sin(4)}{1-2\sin(4)}z \\
 \implies y &= \frac{24\sin(4)}{1-2\sin(4)}0
 \end{aligned}$$

So $y = 0$.

Looking at x with the third equation

$$\begin{aligned}
 x &= 2y + 24z \\
 \implies x &= 2(0) + 24(0) \\
 &= 0
 \end{aligned}$$

We'll look at that $x = y = z = 0$. Our set is NOT linearly dependent! Meaning by definition the system is linearly independent! Recall the set in question

$$\left\{ \begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix}, \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix} \right\}$$

We could have guessed this from the knowledge from part 1 as it is incredibly uncommon for a system to be linearly dependent and still be able to find a solution to the system of linear equations. It is not impossible but it is uncommon so we would expect given that we had a solution in part one, that the system would be linearly independent.